

- 1. Preamble** Students should be familiar with the terms defined in the *Glossary for DI*. Students should show clearly how an algorithm has been applied. Matrix representation will be required but matrix manipulation is not required. Students will be required to model and interpret situations, including cross-checking between models and reality.
- 2. Examination**
- First assessment: June 2019.
 - The assessment is 1 hour and 30 minutes.
 - The assessment is out of 75 marks.
 - Students must answer all questions.
 - Calculators may be used in the examination. Please see *Appendix 6: Use of calculators*.
 - The booklet *Mathematical Formulae and Statistical Tables* will be provided for use in the assessments.
- 3. Notation and formulae** Students will be expected to understand the symbols outlined in *Appendix 7: Notation*.
- Students are expected to know any other formulae that might be required and which are **not** included in the booklet *Mathematical Formulae and Statistical Tables*, which will be provided for use with the paper. Questions will be set in SI units and other units in common usage.
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D1.3 Unit content

What students need to learn:		Guidance
1. Algorithms		
1.1	The general ideas of algorithms and the implementation of an algorithm given by a flow chart or text.	The order of an algorithm is not expected. Whenever finding the middle item of any list, the method defined in the glossary must be used.
1.2	Students should be familiar with bin packing, bubble sort, quick sort, binary search.	When using the quick sort algorithm, the pivot should be chosen as the middle item of the list.